

Bridging the Scale Gap: Cross-View Localization from Dense Rover LiDAR to Coarse Lunar DEMs

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TL;DR Existing LiDAR-to-DEM methods localize only on **distinctive terrain** with **fine (≤ 2 m/px) DEMs** — lunar-scale navigation needs methods **robust to coarse (≥ 5 m/px) DEMs and feature-poor terrain**.

Why LiDAR-to-DEM on the Moon?

- No GNSS · odometry drift · harsh lighting → **no absolute fix**
- **Orbital DEMs** (LOLA / LROC): global prior available before landing
- LiDAR ↔ DEM matching → **absolute, illumination-invariant** pose

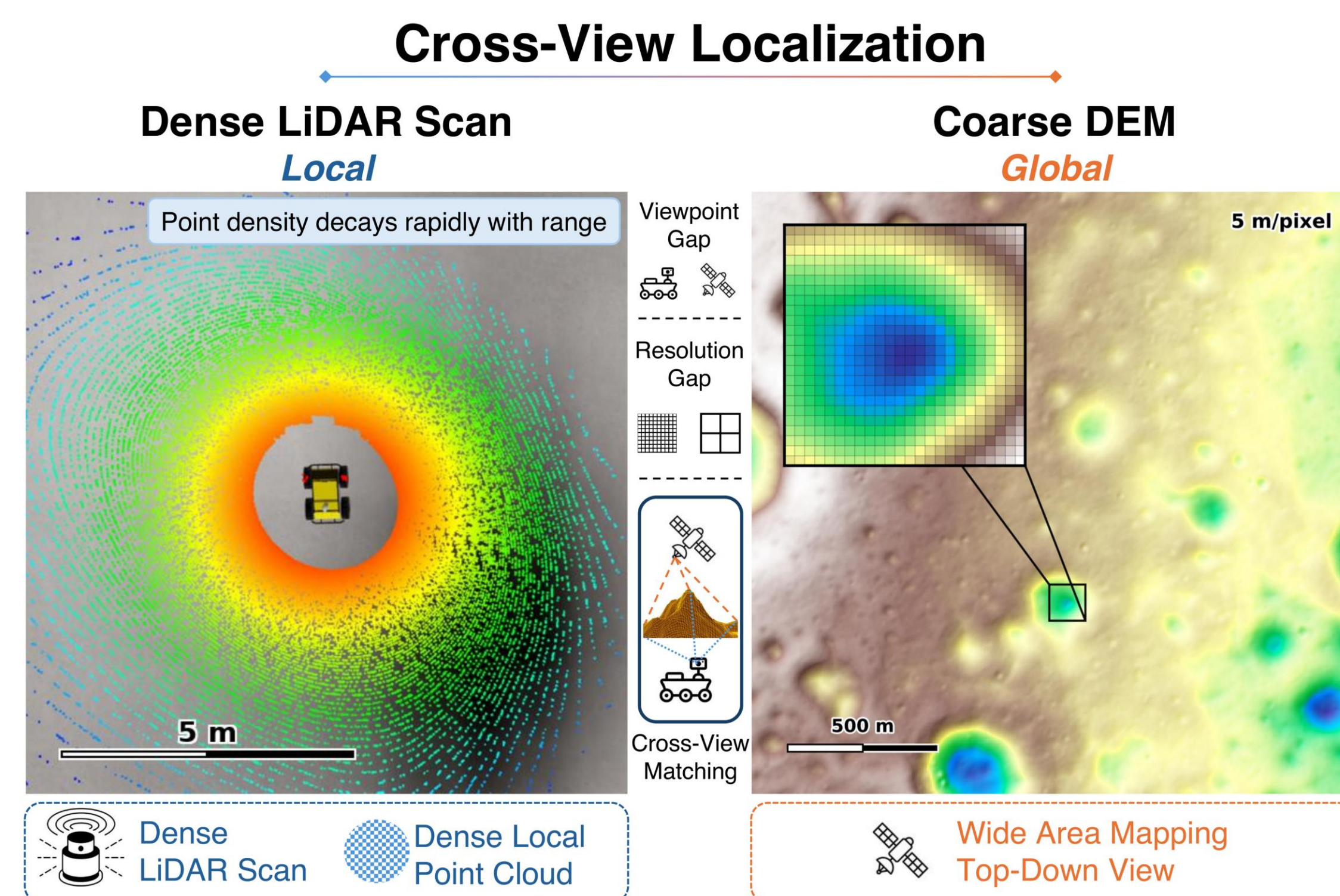


Fig. 1 Dense local LiDAR scan vs. coarse global DEM.

Benchmark

- **Viewpoint gap**: ground-level LiDAR vs. top-down map
- **Resolution gap**: dense scan vs. 1–10 m/px DEM cells
- **Protocol**: original pipelines · single frame · no odometry
- **Setup**: OmniLRS · Husky + Ouster OS1-128 · 5 sites · 6 DEM res. (0.25–10 m/px)
- **Metric**: SR@5m, median / RMSE error

Evaluated Methods

- **DARCES** — planetary, geometric (MOGA core)
- **BEVPlace++** — learned BEV descriptor, lunar fine-tuned
- **RING** — Radon-sinogram descriptor
- **ICP** — multi-start baseline (10 seeds)

Five sites, five morphology classes

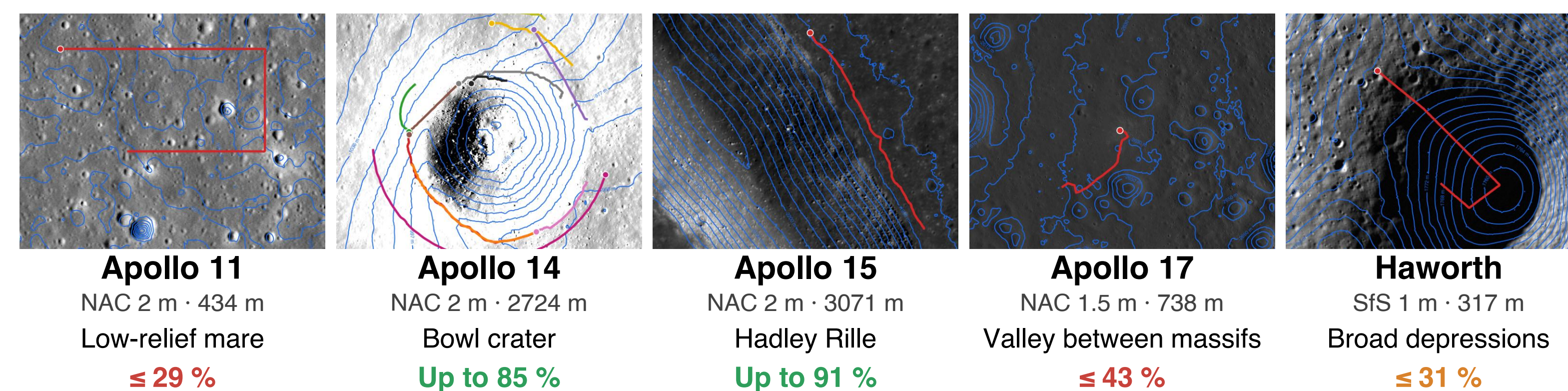


Fig. 2 Sites · rover path (red) · contours (blue).

Localization across DEM resolutions

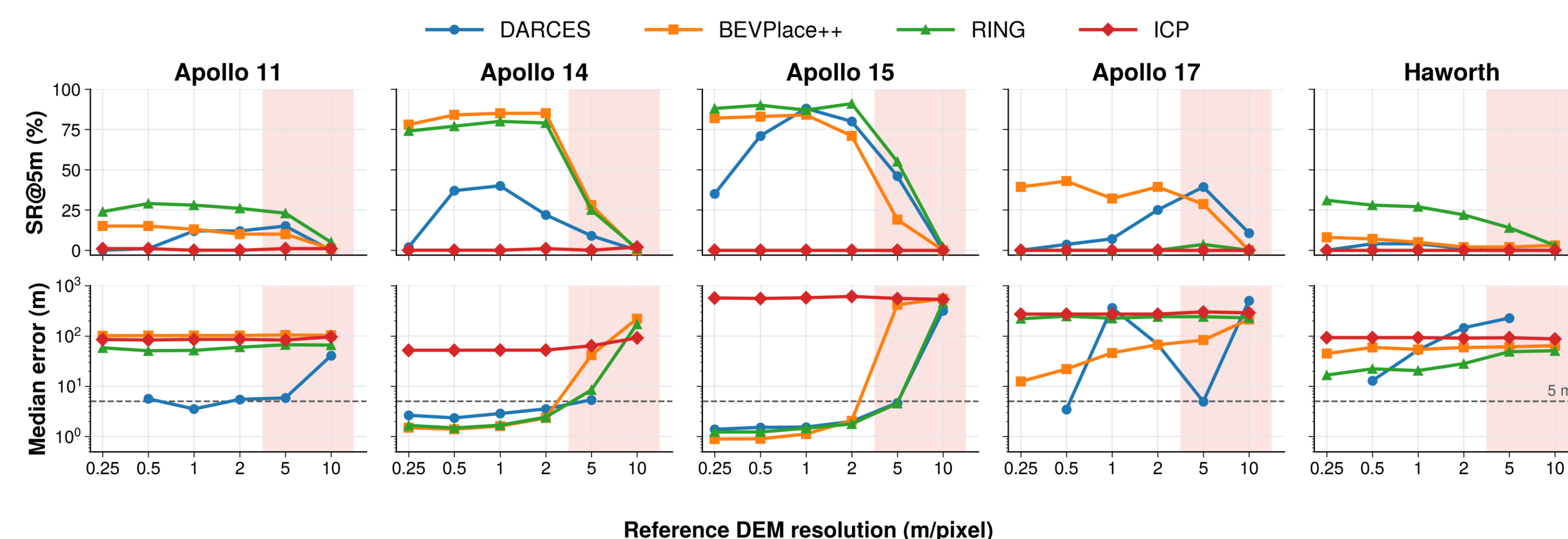


Fig. 3 SR@5m (top) and median error (bottom). Red band: ≥ 5 m/px.

Success vs. failure

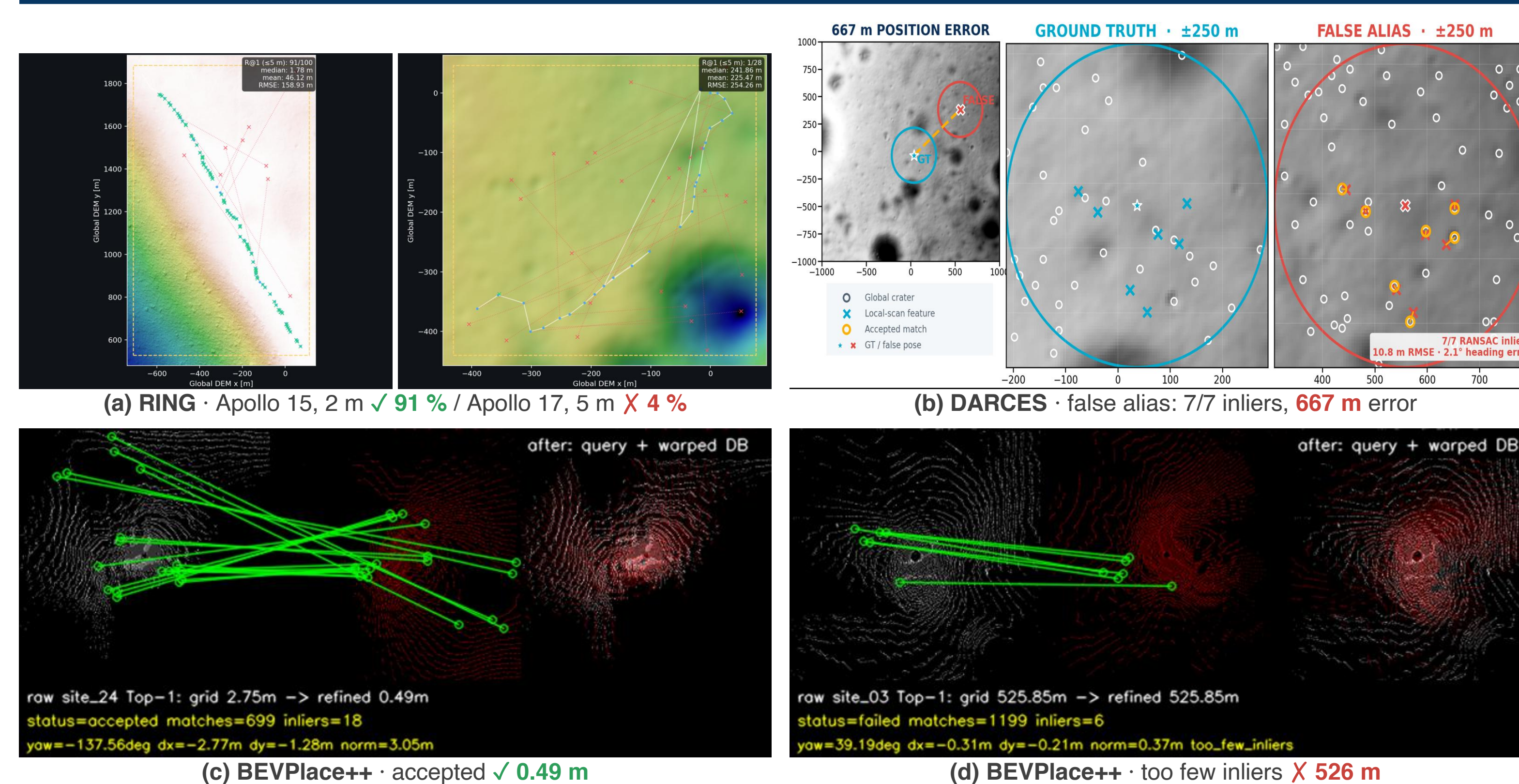


Fig. 4 Single-scan success and failure cases. ICP fails almost everywhere (not shown).

Key findings

- **Terrain > resolution**: crater / rille ≤ 91 % vs. mare ≤ 29 %
- **Cues must be visible locally**: Apollo 17 map-scale relief unused
- **Resolution cliff** (Apollo 14): 85 % → 25 % → 0 % at $\leq 2 / 5 / 10$ m/px
- **Methods**: RING $\geq$ BEVPlace++ except Apollo 17 · DARCES best at mid res. · ICP fails
- **Median $\neq$ reliability**: 1–2 m median, 100–400 m RMSE

Best SR@5m per site

m/px	0.25	0.5	1	2	5	10
Apollo 11	24 R	29 R	28 R	26 R	23 R	5 R
Apollo 14	78 B	84 B	85 B	85 B	28 B	2 I
Apollo 15	88 R	90 R	88 D	91 R	55 R	2 R
Apollo 17	39.3 B	42.9 B	32.1 B	39.3 B	39.3 D	10.7 D
Haworth	31 R	28 R	27 R	22 R	14 R	3 B/R
	finer than orbital		NAC DTM (1–2 m)		most of Moon (≥ 5 m)	

Table 1 Best SR@5m (%) · D/B/R/I = best method.

Takeaway & next steps

- **Bottleneck 1**: coarse (≥ 5 m/px) DEMs — yet they cover most of the Moon
- **Bottleneck 2**: feature-poor terrain — no distinctive craters or rilles
- **Reliability**: outlier rejection / multi-hypothesis checks before deployment
- **Next**: resolution-robust method working **directly on the DEM**
- **Next**: no map-wide pseudo-LiDAR rendering (as RING / BEVPlace++ require)